



MINING DEALER BEST PRACTICE

Disassembly Tool for Servo Piston of 320 and 324D Hydraulic Pumps

Maintenance and
Repair

Application

Component Life
Management

Component
Renewal (CRC)

MARC
Management

Disassembly Tool for Servo Piston of 320 and 324D Hydraulic Pumps		1
1.0	Introduction	2
2.0	Best Practice Description.....	2
3.0	Implementation Steps	3
4.0	Benefits	5
5.0	Resources Required	5
6.0	Supporting Attachments / References	5
7.0	Related Best Practices.....	5
8.0	Acknowledgements	5

DISCLAIMER: The information and potential benefits included in this document are based upon information provided by one or more Cat® dealers, and such dealer(s) opinion of “Best Practices”. Caterpillar makes no representation or warranty about the information contained in this document or the products referenced herein. Caterpillar welcomes additional “Best Practice” recommendations from our dealer network.

August 2023
4.01-230814-1

1.0 Introduction

This tool facilitates the disassembly of the servo piston that controls the angle of the main hydraulic pumps of the 320-324D models and ensures disassembly without damage.

2.0 Best Practice Description

The servo piston needs to be removed for rebuild. The servo piston is not a fixed part. Therefore, it moves left and right during the removal. These movements can cause the servo piston pin to break during removal.

In addition, it was difficult to tighten according to the test procedures [SIS 2.0 \(cat.com\)](#) during the assembly phase. The breakage of the servo piston pin is a high additional cost because the broken pin does not have a single sale. If the pin breaks, the piston, and pins must be purchased and replaced.

A tool has been designed to prevent the pin from breaking and to facilitate the removal of the servo piston. The tool is placed around the servo piston during removal and prevents the piston from moving in either direction. The tool allows removal without breaking the pin. The piston that can be removed without damage is shown in the figure below.

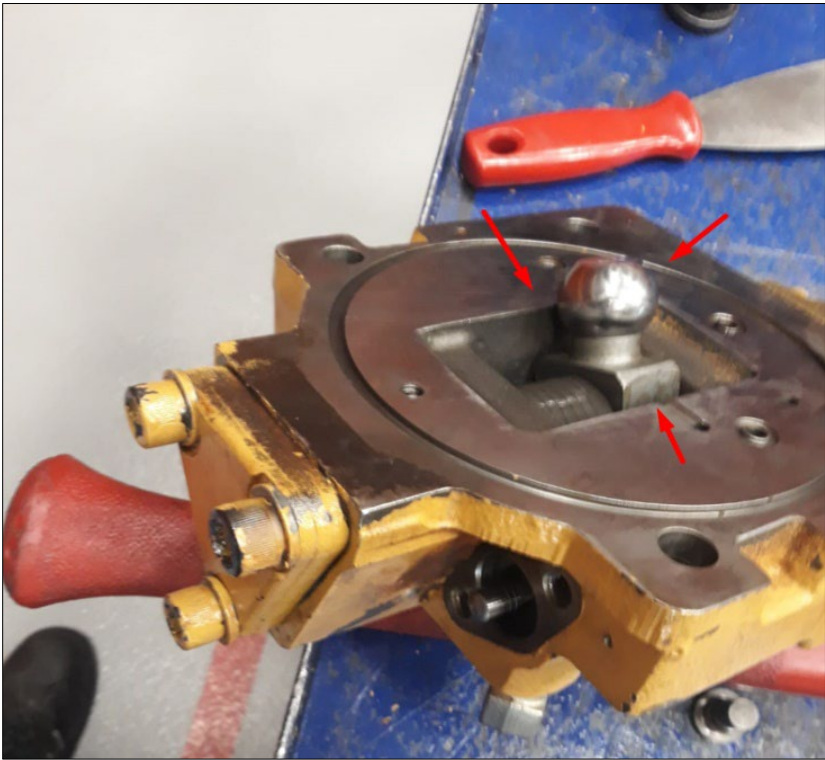


Disassembly support pin assembly without damage the fixing pin.

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Disassembly Tool for Servo Piston of 320 and 324D Hydraulic Pumps	DATE 8/14/2023	CHG NO 00	NUMBER 4.01-230814-1

3.0 Implementation Steps

Step 1:
The piston in the image must be fixed because it is unstable and moves to its right-left directions. The pin under the servo piston engages in the fixing slot. The pin is connected to the servo piston. When the servo piston moves, it forces the pin to move and begins to operate in the fixing split. Because the fixing split is stronger than the pin, it breaks the pin during assembly and disassembly. The tool holds the head of the servo piston in place and prevents movement during assembly/disassembly.



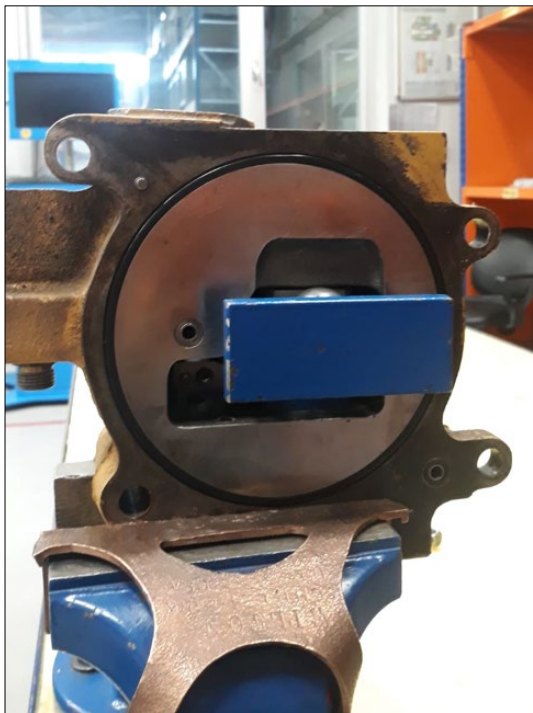
PIN
FIXING FORKE



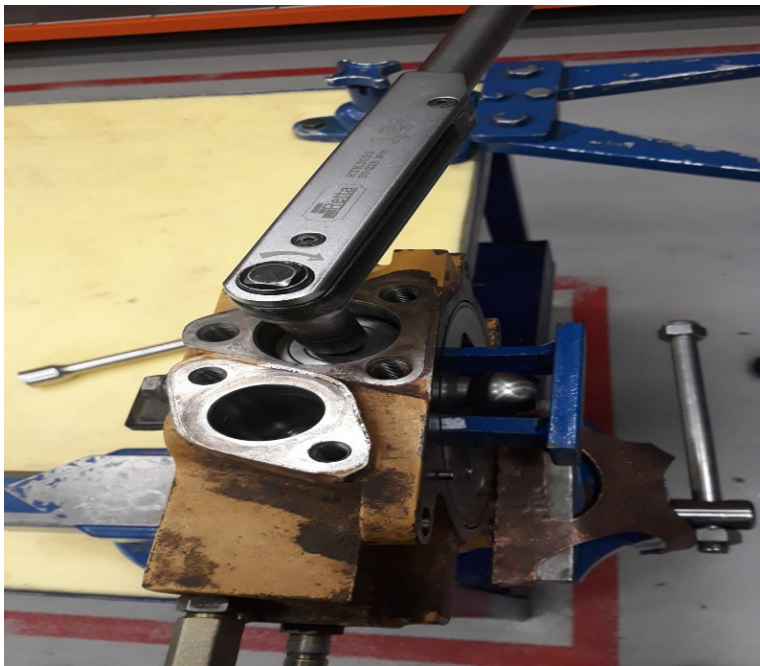
INTERLOCKED

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Disassembly Tool for Servo Piston of 320 and 324D Hydraulic Pumps	DATE 8/14/2023	CHG NO 00	NUMBER 4.01-230814-1

Step 2:
The tool is fixed around the piston.



Step 3:
The piston is fixed, and assembly processes can start.



THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR

Disassembly Tool for Servo Piston of 320 and 324D
Hydraulic Pumps

DATE
8/14/2023

CHG
NO
00

NUMBER
4.01-230814-1

4.0 Benefits

- Fast operation
- Removing the replacement part without damaging it
- Prevention extra costs
- Customer satisfaction

5.0 Resources Required

The attachment tool was produced in our own workshop. Our existing materials are used for production. A person working in the welding room finished production after 4 hours of work.

Item	Requirements
Material	~100 €
Labor	1 worker x 4 hours

6.0 Supporting Attachments / References

- Drawing in the appendix
- i05916114 Disassembly and Assembly [SIS 2.0 \(cat.com\)](#)

7.0 Related Best Practices

None

8.0 Acknowledgements

This Best Practice was written by:

Esra Ulubay
CRC Planning Officer
Borusan Makina ve Güç Sistemleri

This Best Practice designed by:

Abdullah Akbenli
Expert Technician
Borusan Makina ve Güç Sistemleri

Project and Best practice supported by:

H.Yağız Vural
Workshop Engineer
Borusan Makina ve Güç Sistemleri

THE INFORMATION HEREIN MAY NOT BE COPIED OR TRANSMITTED TO OTHERS WITHOUT THE WRITTEN PERMISSION OF CATERPILLAR			
Disassembly Tool for Servo Piston of 320 and 324D Hydraulic Pumps	DATE 8/14/2023	CHG NO 00	NUMBER 4.01-230814-1